```
import java.util.Random;
import java.util.Arrays;

public class MergeSort {

    // Merge Sort algorithm
    public static void mergeSort(int[] arr, int left, int right) {

        if (left < right) {

            int mid = left + (right - left) / 2;

            // Sort left half
            mergeSort(arr, left, mid);

            // Sort right half
            mergeSort(arr, mid + 1, right);

            // Merge both sorted halves
            merge(arr, left, mid, right);
        }
    }

    // Merge two sorted parts
    public static void merge(int[] arr, int left, int mid, int right) {

        int n1 = mid - left + 1;
        int n2 = right - mid;

        int[] leftArray = new int[n1];
        int[] rightArray = new int[n2];

        // Copy values into temporary arrays
        for (int i = 0; i < n1; i++) {
            leftArray[i] = arr[left + i];
        }

        for (int j = 0; j < n2; j++) {
            rightArray[j] = arr[mid + 1 + j];
        }

        int i = 0;
        int j = 0;
        int k = left;

        // Compare and merge values in sorted order
        while (i < n1 && j < n2) {

            if (leftArray[i] <= rightArray[j]) {
                arr[k] = leftArray[i];
                i++;
            } else {
                arr[k] = rightArray[j];
                j++;
            }

            k++;
        }
```

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        // Copy remaining values from left array
        while (i < n1) {
            arr[k] = leftArray[i];
            i++;
            k++;
        }

        // Copy remaining values from right array
        while (j < n2) {
            arr[k] = rightArray[j];
            j++;
            k++;
        }
    }

    // Main method: run Merge Sort independently
    public static void main(String[] args) {

        int size = 1000;
        int[] arr = new int[size];

        Random random = new Random();

        // Generate random numbers
        for (int i = 0; i < size; i++) {
            arr[i] = random.nextInt(10000);
        }

        System.out.println("===== MERGE SORT =====");
        System.out.println("Array Size: " + size);

        System.out.println("\nArray Before Sorting:");
        System.out.println(Arrays.toString(arr));

        long startTime = System.nanoTime();

        mergeSort(arr, 0, arr.length - 1);

        long endTime = System.nanoTime();

        System.out.println("\nArray After Sorting:");
        System.out.println(Arrays.toString(arr));

        long executionTime = endTime - startTime;

        System.out.println("\nExecution Time: "
                + executionTime + " nanoseconds");

        System.out.println("Execution Time: "
                + executionTime / 1_000_000.0 + " milliseconds");
    }
}
```

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